



Intestinal Neoplasms & Diverticulum

Pathology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.**EASY**

FAP is caused by mutation of which gene?

- A. BRCA2
- B. APC
- C. MLH1
- D. TP53
- E. KRAS

2.**EASY**

What is the minimum polyp count required for a diagnosis of classic FAP?

- A. 5
- B. 500
- C. 100
- D. 10
- E. 20

3.**EASY**

The vast majority of colorectal carcinomas are:

- A. Squamous cell carcinoma
- B. Neuroendocrine tumor
- C. Lymphoma
- D. Adenocarcinoma
- E. GIST

4.**EASY**

Which is the commonest site of acquired diverticula?

- A. Sigmoid colon
- B. Duodenum
- C. Ascending colon
- D. Jejunum
- E. Ileum

5.**MEDIUM**

GISTs are believed to arise from which cell type?

- A. Neural crest cells
- B. Paneth cells
- C. Interstitial cells of Cajal
- D. Enterochromaffin cells
- E. Smooth muscle cells of the muscularis mucosae

6.**MEDIUM**

Which site is most commonly affected by GIST?

- A. Stomach (~60%)
- B. Small bowel (~60%)
- C. Esophagus (~60%)
- D. Colon (~60%)
- E. Rectum (~60%)

7.

MEDIUM CASE

A colonoscopy in a 68-year-old shows a circumferential, infiltrating stricture in the descending colon described as a "napkin-ring" lesion.

This presentation is typical of carcinoma of the:

- A. Appendix
- B. Left colon
- C. Right colon
- D. Duodenum
- E. Rectum

8.

MEDIUM

Which dietary factor enhances hepatic synthesis of cholesterol and bile acids that bacteria can convert into carcinogens?

- A. High fiber intake
- B. High fat intake
- C. Low fat intake
- D. Vitamin E supplementation
- E. Low refined carbohydrate intake

9.

MEDIUM

Aspirin and other NSAIDs are thought to protect against colorectal carcinoma primarily via inhibition of which enzyme?

- A. Cytochrome P450
- B. Lipoxygenase
- C. COX-2
- D. Xanthine oxidase
- E. COX-1

10.

MEDIUM CASE

A patient with an ileal neuroendocrine tumor is found to have elevated urinary 5-HIAA.

5-HIAA is the hepatic breakdown product of which substance secreted by the tumor?

- A. Insulin
- B. Gastrin
- C. Somatostatin
- D. Vasoactive intestinal peptide
- E. Serotonin (5-HT)

11.

MEDIUM

Appendiceal neuroendocrine tumors, compared with ileal NETs, tend to behave:

- A. More benignly
- B. More aggressively
- C. With higher rates of carcinoid syndrome
- D. With higher metastatic potential
- E. Identically

12.

MEDIUM

A congenital diverticulum, by definition, is lined by:

- A. Muscularis propria only
- B. No epithelial lining
- C. Serosa only
- D. All layers of the gut wall
- E. Mucosa and submucosa only

13.

MEDIUM

Which of the following is part of the "rule of 2's" for Meckel diverticulum?

- A. Three times more common in females
- B. Located within 2 feet of the ileocecal valve
- C. Most often symptomatic by age 10
- D. Approximately 10 cm long
- E. Occurs in 10% of the population

14.

MEDIUM

Which factor is used, together with size and mitotic rate, to predict the behavior of a GIST?

- A. Presence of ulceration
- B. Patient age
- C. Serum CEA level
- D. Tumor color
- E. Anatomic location

15.

HARD CASE

A resected colonic mass shows malignant glands with marked central necrosis infiltrating into the muscularis propria.

This histologic pattern is most consistent with:

- A. Colorectal adenocarcinoma
- B. Meckel diverticulum
- C. Hirschsprung disease
- D. GIST
- E. Neuroendocrine tumor

16.

HARD

Which route of CRC spread occurs first to the liver, given the venous drainage of the bowel?

- A. Direct spread through the wall
- B. Transcoelomic spread
- C. Blood-borne spread via the portal vein
- D. Lymphatic spread to mesenteric nodes
- E. Retrograde venous spread

17.

HARD CASE

A 45-year-old with a midgut neuroendocrine tumor and known liver metastases develops flushing, diarrhea, and new tricuspid stenosis on echocardiogram.

Why does carcinoid syndrome typically require the presence of liver metastases?

- A. The liver is the only site capable of producing serotonin
- B. Liver metastases convert the tumor to an adenocarcinoma
- C. Liver metastases allow serotonin to bypass hepatic inactivation and enter the systemic circulation
- D. The primary tumor cannot produce serotonin without hepatic support
- E. Liver metastases increase gastrin secretion

18.

HARD

A GIST is classified as high-risk based on which combination of parameters?

- A. Age of the patient and tumor color
- B. Serum 5-HIAA level
- C. Presence of ectopic gastric mucosa
- D. Large size, unfavorable location, and high mitotic rate
- E. Site of the primary tumor alone

19.

HARD CASE

A 3-year-old presents with painless lower GI bleeding. Imaging and surgical resection reveal a true diverticulum arising from the antimesenteric border of the distal ileum, containing patches of ectopic tissue.

This ectopic tissue most likely explains the bleeding because it is:

- A. Renal tissue causing hematuria misidentified as GI bleeding
- B. Acid-secreting gastric mucosa causing peptic ulceration of adjacent ileal mucosa
- C. Squamous epithelium prone to keratinization
- D. Thyroid tissue causing hyperthyroid-related bleeding
- E. Hepatic tissue prone to hemorrhage

20.

HARD

Which statement correctly distinguishes acquired sigmoid diverticula from Meckel diverticulum?

- A. Meckel diverticulum lacks muscularis propria, while acquired diverticula have all wall layers
- B. Both are true diverticula with all wall layers
- C. Both lack all wall layers
- D. Acquired diverticula are more common in the ileum than the sigmoid colon
- E. Acquired diverticula lack or have attenuated muscularis propria, while Meckel diverticulum has all wall layers

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	FAP results from mutation of the APC tumor suppressor gene on chromosome 5.
2	C	At least 100 colorectal adenomatous polyps are required for a diagnosis of classic FAP.
3	D	The vast majority of colorectal carcinomas are adenocarcinomas.
4	A	The sigmoid colon is the most common site of acquired diverticula.
5	C	GIST is believed to arise from the interstitial cells of Cajal, the GIT pacemaker cells.
6	A	The stomach accounts for about 60% of GISTs, followed by the small bowel (~30%).
7	B	Left colon carcinoma classically presents as a circumferential, infiltrating napkin-ring constriction causing bleeding, altered bowel habit, and obstruction.
8	B	High fat intake enhances hepatic cholesterol and bile acid synthesis, which intestinal bacteria can convert into carcinogens.
9	C	NSAIDs are thought to be protective via inhibition of COX-2, which normally produces prostaglandin E2 and promotes epithelial proliferation.
10	E	Midgut NETs commonly secrete serotonin, which is inactivated by the liver to 5-HIAA; elevated urinary 5-HIAA aids diagnosis.
11	A	Appendiceal NETs tend to behave in a benign manner, whereas ileal NETs are low-grade malignant and can metastasize to nodes or liver.
12	D	Congenital (true) diverticula are lined by all layers of the gut wall, unlike acquired diverticula which lack or have attenuated muscularis propria.
13	B	Meckel diverticulum is typically located within 2 feet (60 cm) of the ileocecal valve, per the rule of 2's.
14	E	The most important parameters for predicting GIST behavior are size, anatomic location, and mitotic rate.
15	A	Adenocarcinoma shows malignant cells in a glandular pattern with marked necrosis and infiltration of the muscle layer.
16	C	Blood-borne metastasis to the liver occurs via the portal vein, which drains the bowel directly to the liver.
17	C	Portal-vein serotonin is normally inactivated by the liver to 5-HIAA before reaching the systemic circulation; liver metastases release serotonin directly into the systemic circulation, producing carcinoid syndrome.
18	D	GIST behavior/risk is predicted using tumor size, anatomic location, and mitotic rate.
19	B	Ectopic gastric mucosa in Meckel diverticulum secretes acid, which can ulcerate adjacent normal ileal mucosa and cause bleeding.
20	E	Acquired (sigmoid) diverticula lack or have attenuated muscularis propria (false diverticula), while Meckel diverticulum is a true diverticulum with all four wall layers.

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